

Satellite access services

J F Wakeling and W H Dobbie

Satellite systems are unique in that they can effectively provide access, trunk and international links through the same physical infrastructure due to their inherent wide-area coverage. Satellites are increasingly being seen as a flexible delivery mechanism which can meet a diverse set of needs from Internet access to global packet-based private networks. This paper analyses the access capabilities of existing satellite systems in terms of capabilities and limitations and looks at future system proposals which aim to fully integrate satellite systems into terrestrial networks for the delivery of advanced services. The long-term objective is to give service providers an additional access technology option for both national mass markets and global connectivity.

1. Introduction

Satellite systems offer a unique delivery platform in that they can effectively provide access, trunk and international links through the same physical infrastructure due to their inherent wide coverage characteristics. This is particularly true of the current generation of geostationary (GEO) satellites which can provide coverage of about a third of the world from one spacecraft. The first use of satellites was the provision of international telephony services, but they have evolved to serve many diverse markets in both broadcast and interactive applications.

One major use is for broadcast-related services. These range from provision of point-to-point contribution links including those from mobile and transportable terminals to satellite broadcasting including business TV and/or data delivery and satellite Internet delivery. All of these services are now benefiting from the development of digital TV multiplexing and encoding technology and particularly from the standardisation which has been achieved by the 230 plus companies involved with the European Digital Video Broadcasting (DVB) project. Satellite pay TV broadcasting has already had a major impact on the development of the residential communications market. This paper discusses the ways in which digital technology will be used to increase the number of channels and enhance the quality (when the display technology justifies it) plus the addition of interactivity via the PSTN later this year and of 'push-Internet' services in the future.

Very small aperture terminals (VSATs) have been used for closed private networks for many years, both in national and international applications. This paper identifies the issues that have stopped them being used for mass public access up to now. The proposed GEO broadband satellite

systems such as Spaceway and Astrolink are discussed in terms of their enhanced multiplexing capabilities and ability to reduce cost-per-bit, which may enable them to address wider markets than existing VSAT services. Proposals for non-GEO broadband systems, such as Teledesic and SkyBridge, are reviewed in terms of their specific network and service management challenges. A similar review of mobile satellite services is given, including a discussion of the emerging non-GEO systems, such as ICO and Iridium.

The paper closes with conclusions on the applicability of satellite services to the UK access market and the opportunities that they present to international access provision.

2. Broadcast related services

2.1 TV distribution

Point-to-point GEO satellite links have been used for international distribution of TV programmes since the 1960s. Most people will have seen archive footage of the Beatles contribution to the first global TV event. This was a precursor to Live Aid and other more recent events which were viewed live in almost every country in the world. There is no doubt that the extra audiences which such communications links provided was a major factor in the development of TV which contributed to its significant impact on society.

The early services required very large dishes at both ends of the link but over the last 15 years it has become common to use mobile uplink terminals on trucks for coverage of sporting and other special events and very small

transportable (fly-away) uplink terminals for news in war zones and disaster areas. The mobile terminals are able to transmit full broadcast-quality material and even HDTV but the transportable terminals are often limited by the available power and the dish size to low-quality video and audio. Both types of terminals are normally equipped with at least one independent two-way satellite audio engineering channel to help with line-up and the inevitable problems which occur. The Inmarsat network described in section 4.1 is sometimes used for these channels.

In the past the video and audio signals were all analogue, but digital coding is now becoming normal. The DVB project [1] which has defined specifications for most aspects of digital satellite broadcasting, as discussed below, is currently defining a specification for mobile and transportable digital satellite news-gathering terminals. This will include the option of using 8PSK with trellis-coded modulation to maximise the efficiency of the link and will allow for up to four engineering channels to be provided via the same satellite capacity used for the TV service.

2.2 Interactive broadcast TV

Satellite broadcasting has also had a significant effect on our lives, mainly through the introduction of extra TV channels competing with traditional terrestrial TV. Within Europe these were initially delivered to large dishes which fed cable TV networks using frequencies in the Ku band (12—14 GHz) allocated for communications. As satellite and receiver technology developed it became possible to use smaller and smaller receive dishes and a dedicated broadcasting band was allocated by the ITU in 1977. This assumed the use of high power GEO satellite transponders in association with 90 cm receive dishes. During the mid-1980s, however, it became practical to receive analogue PAL TV signals from medium-power satellites using a 60 cm dish and this, in conjunction with the introduction of video scrambling, plus conditional access control, enabled the commercialisation of direct satellite pay TV broadcasting to the home.

The multichannel pay TV environment which has resulted from the introduction of cable TV networks (partly supported by cable telephony revenues) and direct-to-home satellite TV services has, to a great extent, led to the development of low bit rate digital coding and multiplexing systems based on the moving picture experts group (MPEG) specifications. Initially these were implemented in a proprietary manner but over recent years the DVB project has developed a set of specifications covering many aspects of such systems necessary to allow compatibility [1, 2]. All of the digital systems that are being developed and introduced within Europe (and other areas of the world, particularly where PAL is established) are therefore now based on the DVB specifications.

The satellite modulation and coding specification is based on QPSK with concatenated Reed Solomon and Viterbi forward error protection coding (FEC) plus interleaving. A range of symbol rates and Viterbi coding rates are defined, allowing an optimal trade-off to be made for a particular satellite transponder. This also makes it possible to trade dish size against data rate. The payload is used to carry an MPEG-2 multiplex which allows a mixture of audio, video and data services to be combined in a time division multiplex. The video coding rate can be allocated in a flexible manner which means that different levels of picture quality can be provided for different applications. Service information tables are used to describe the different services so that set-top boxes can extract and decode all of the elements needed for a particular service selected by the user. A common scrambling algorithm is defined and it can be used in conjunction with different proprietary conditional access (CA) control systems. This means that the same service can be sold to subscribers with set-top boxes equipped with different CA processors. A common interface has also been specified and if this is fitted it is possible for a single set-top box to include more than one CA module.

The flexibility offered by low bit rate MPEG video coding allows each transponder to deliver many more TV channels than with analogue. A typical 33 MHz transponder can deliver 34 Mbit/s to a 45—50 cm dish using $2/3$ rate Viterbi FEC. The 31.7 Mbit/s remaining after demultiplexing can be used to carry 8×4 Mbit/s TV channels with quality similar to that for current analogue terrestrial TV services. Alternatively, it could deliver 16 channels with quality as for the BT ADSL/VOD trials which is now accepted as adequate for a wide range of applications.

This significant increase in bandwidth efficiency allows the broadcaster to increase the number of channels which they can deliver. BSkyB plans to launch a 200-channel pay TV package in 1998 which will include many new niche channels and near video-on-demand for top movies. They plan to use a new family of satellites located at 28.2 degrees east which will be provided by SES/Astra. These will use the whole of the Ku-band including the broad-casting frequency allocations. Eutelsat and others are also planning similar families of collocated satellites.

In the future it is also planned to add satellites operating in the Ka-band (20—30 GHz) even though the severe rain fades which occur at these higher frequencies may make it necessary to use larger dishes. This may make it desirable to use a different form of videocoding where the basic picture is more heavily protected than the enhancement information, allowing the picture to degrade progressively rather than just fail. This approach could help to enable the introduction of HDTV services to wide-aspect ratio large-

screen displays when these become available at an acceptable price.

DVB has also specified a number of different options for implementation of return channels for interactive TV services plus a network-independent protocol stack [2]. The most practical options are probably the PSTN or the ISDN in the near term, but DVB has also defined solutions based on DECT and GSM. Work is also in progress aimed at defining a Ku-band satellite return channel. This would require the addition of a transmitter and combiner to the customer's dish and considerable work will be required to implement this at a low enough cost to make it practical. It will also be necessary to ensure that incorrect installation and use do not result in spectral pollution. Nevertheless, SES/Astra and others are convinced that it will be feasible to deliver low bit rate messages back to a service provider via such channels. SES are currently procuring satellites with Ka-band transponders and plan to introduce pilot services to business customers in 1999.

The BSkyB services will be complemented shortly afterwards by interactive services from the BIB consortium which consists of BSkyB, BT, Matsushita and the Midland Bank. Text, graphics, still pictures and video material will be provided by satellite and used in conjunction with a PSTN return channel for shopping, information and other applications. A customer will be able to browse through a range of broadcast multimedia channels and then in the case of a shopping application use the interaction channel to find out more about a particular product and then pay for it [3]. The BIB services will not in fact use the DVB specifications for interactivity because they will be based on the proprietary Open TV applications programming interface (API) selected by BSkyB.

DVB is currently defining an open standard API to ensure that in the future set-top boxes need not be specific to one service provider. This will allow for broadband interactive services and Internet. DVB has also almost completed a specification for an ATM-based physical interface which will allow boxes to be connected to telco network termination units enabling delivery of broadband interactive TV services via xDSL or fibre networks.

2.3 *Business TV and data*

The low-cost set-top boxes and systems which have been developed for interactive digital broadcast TV will also be used for business applications such as secure delivery of corporate training or company-wide news services. Rather than using a complete satellite transponder it has become possible to use a small part of the total payload which reduces costs and encourages more use. The boxes are also flexible enough to allow data download to PCs and other devices through suitable interfaces. PSTN or other interactive channels can also be used where necessary.

These one-way VSAT services can be used to update local databases (e.g. with information on stolen credit cards), deliver MUSAC to stores or to download new software, etc. They become extremely cost effective when a company has a large number of physically diverse sites, as for instance with chains of petrol stations or car dealerships. BT has provided these types of network for many years and the advent of cheap and flexible set-top boxes and systems is likely to increase demand.

2.4 *Satellite Internet*

The delivery of push-Internet data represents a logical extension of the use of satellites. Data is normally transmitted as part of a carousel which is continuously repeated. Authorised customers are able to decode the broadcast (or multicast) data and either use it immediately or cache it (normally on a PC hard disk) for future consumption completely on-demand. The main limits are the size of the carousel that is transmitted, its repetition rate (which depends on the amount of capacity used) and the size of the local cache. Video-rich services can be included providing there is a large enough customer base to justify their inclusion on the carousel. It is also possible to transmit push-Internet services according to a linear schedule. This is similar to digital TV except that the content is formatted using HTML for decoding in a PC or a network computer.

It is also possible to deliver fast pull-Internet (often termed turbo-Internet) to a particular customer following a request, normally via the PSTN. TCP/IP is used instead of UDP/IP (which is used for push-Internet) to ensure that the data is received without errors. The necessary acknowledgement messages are transmitted via the PSTN link which means that the data rate over the satellite channel is restricted. Nevertheless, providing TCP/IP extensions are used to deal with the window sizes (especially during start-up), it is possible to increase the forward data rate to 10 times that of the PSTN link or more. The main limit here is, however, the cost of the satellite capacity which is relatively high for such mass-market point-to-point applications. This is likely to ensure that fast pull-Internet is mainly used for business services at least until Ka-band capacity becomes readily available. Even then it is unlikely that there will be enough capacity to support video-rich pull-Internet services.

The first satellite Internet system was introduced by Hughes/Direct PC in the USA. It was based on fast pull delivery initially and proprietary hardware. It was therefore relatively expensive which limited its take-up. DVB has recently defined a specification allowing IP data to be tunnelled via the MPEG streams. This means that the mass-market technology developed for satellite broadcast TV can be used. It is only necessary to add suitable filters to extract the IP data. A number of companies including Hughes, Media 4 and Philips have now developed systems including relatively cheap PC cards based on the DVB specifications.

It is also possible to interconnect with a LAN and associated local caches allowing access without a dedicated PC card.

Satellite Internet services based on this technology are now being introduced in Europe by ESM (a consortium including SES, Deutsche Telekom and Intel) and others. The Fantastic Corporation (TFC) is providing a push-Internet service via the ESM system and this is generating considerable interest from potential service providers.

BT is currently providing accommodation for the required routers and multiplexers plus an uplink for a major trial funded by Eutelsat and is also likely to participate as a service provider. Satellite delivery of Internet could provide an efficient and expedient means of delivering services to complement the more conventional BT network-based Internet offerings in the near term. Participation in this trial will allow the benefits and potential problems to be identified and the capabilities to be demonstrated.

3. Fixed communications services

3.1 Current VSAT systems

Very small aperture terminals (VSATs) have been used to provide private network connectivity across vast areas for over 20 years. The use of VSATs was pioneered in the USA where they were used to provide multiple site connectivity for large corporations in preference to terrestrial private circuits several thousand miles long. There are now more than 150 000 business VSAT terminals in operation in the USA, mostly used by the retail sector for credit card verification and inventory tracking. Take-up of VSAT services across Europe had not been as great as in the USA, due to the smaller distances involved, the relative maturity of the long-distance networks in western Europe and the relatively restrictive national regulatory regimes. Deregulation across Europe, although slow, and the opening up of eastern European markets has led to VSAT market growth outstripping that in the USA by a factor of two. Developing regions such as Africa, Asia and the Pacific Rim are growth markets for VSATs, although restrictive regulation is a barrier to VSAT services achieving their full potential. There are currently more than 250 000 VSAT terminals deployed world-wide.

Typical VSAT applications include remote LAN access for single users, LAN-LAN interconnect, business TV, data and video distribution and broadcast, client/server transactions, and intranet and Internet access. VSAT systems can be provided in point-to-point, star or mesh configurations as indicated by the topologies in Fig 1.

The satellites in use today are all transparent, that is to say there is no switching function performed on the spacecraft itself. VSAT systems can provide routing between terminals on a per-call basis, but the satellite itself

only provides the transmission medium. Most USA VSAT systems use the C-band (4–6 GHz) which has a very benign propagation environment which leads to high service availability at the expense of relatively large user antennas of 2–3 m diameter. European VSAT systems use a section of the Ku-band (12–14 GHz) due to spectrum congestion in the C-band and the fact that the physics of the higher frequency allows smaller user antennas (1.8–2.4 m) to be used for the same system performance.

Typical VSAT system components are identified in Fig 2. The system is really quite simple, with the key remote site equipment being the VSAT antenna. Satellite systems use line of sight radio links and antenna placement is crucial to the quality of the link. Partial obstruction of the line-of-sight path or signal reflections from nearby clutter can cause unacceptable interference. There is a limit of about 100 m on the distance between the RF head on the antenna and the modem before signal loss becomes an issue. These issues impact on VSAT installations in urban areas, tall buildings and shared accommodation (e.g. flats, office blocks).

For closed VSAT networks the hub usually provides connectivity to a host computer or LAN on customer premises via a leased line. Applications can be multiplexed at the remote site and demultiplexed at the host if multiple applications using different physical host networks are to be supported, e.g. voice and inventory tracking. VSAT systems typically support X.25 and FR carrying higher layer data protocols such as DVB, MPEG, TCP, UDP, PPP and IP-based applications. It has also been demonstrated that VSAT systems can support ATM bearer services without problems [4].

Technically, there is no reason why VSAT systems cannot be used to provide simple access services to residential customers as well as business users. Satellite systems can support applications based on the same technologies that will be deployed on terrestrial networks, including ATM, FR and TCP/IP. There is a throughput limitation on interactive TCP/IP-based applications of about 900 kbit/s because of the finite window size available with the basic TCP stack, but this is more than adequate for the majority of today's applications. It will become a constraint should interactive full-motion video become the application of choice, but this is not expected to be the case for some time.

The key barrier to wider VSAT use today is cost. VSAT system costs are currently driven by the following three factors.

- System licence costs

In the UK, for example, a DTI Radio Communications Agency VSAT licence to transmit a 64 kbit/s link costs

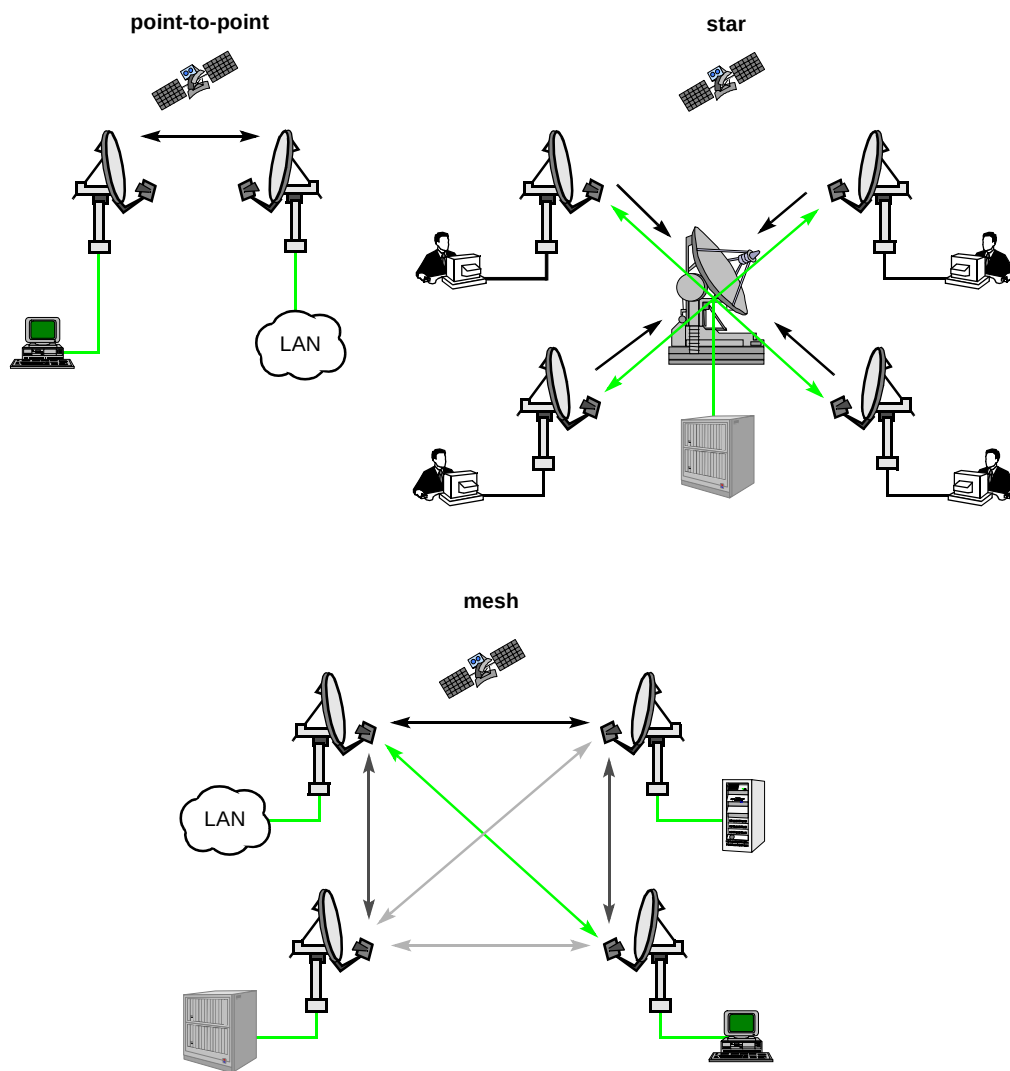


Fig 1 Generic VSAT network configurations.

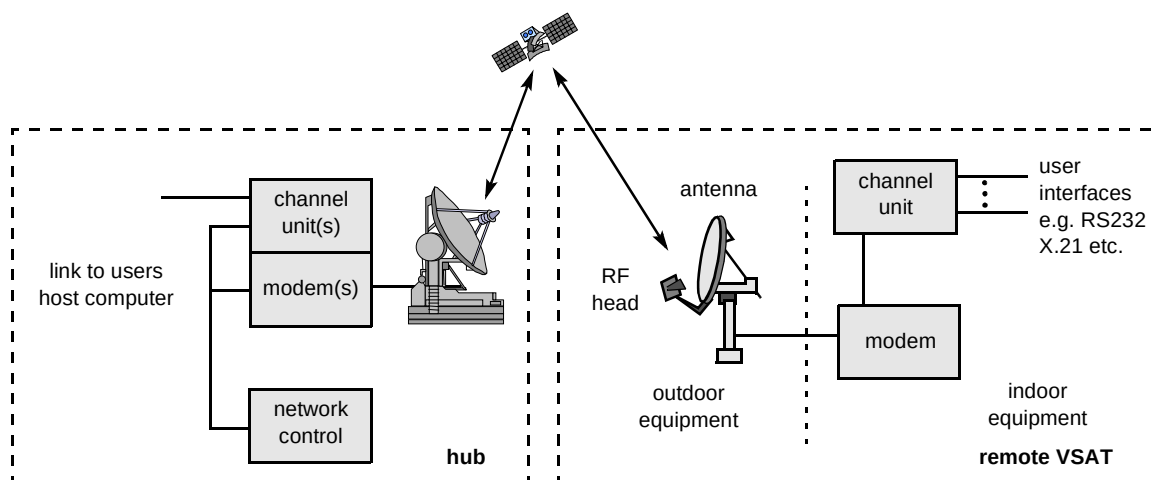


Fig 2 Basic VSAT system components.

between £40—£2000 depending on the number of VSATs in the network [5]. Similar licence charges are made by national administrations across Europe. In

other parts of the world licence costs can be completely prohibitive to service deployment or licences may simply not be available.

- Space segment costs

Typical communications satellite capacity costs vary with coverage, power and bandwidth requirements. The cost of a Ku-band simplex 64 kbit/s channel operating to a VSAT in Europe costs in the region of \$5k—10k per year at current tariffs.

- Equipment costs

Basic VSAT terminals retail at around £5k, not including the DTE. A gateway station providing access to a host network, either a private or public network, costs in the region of £500k—£2M, depending on the size and complexity of the network.

These costs combine to make even a simple VSAT link an expensive item. Hub and space segment costs can be shared but at the cost of additional complexity, and may result in trading off grade and quality of service. The application to be supported has a major impact on the viability of a VSAT solution. It is much easier to multiplex short transaction traffic, such as credit card verification, than it is to combine voice traffic while maintaining a high level of GoS at minimal cost. However, the ability of VSAT systems to connect remote or widely dispersed sites makes them attractive as a single technology solution for multinational corporations (MNCs) who require a single, flexible, easily managed network. VSATs can also be used to provide expedient access solutions for remote sites prior to the roll-out of a terrestrial link in order to complete a private network. Finally, VSATs can also be used as back-haul routes for access traffic which is served by other technologies, e.g. wireless local loop.

3.2 Future broadband systems

Several suppliers have proposed a new generation of broadband satellite systems aimed at reducing the cost of both space segment and user equipment by:

- multiplexing traffic on-board the satellite at the link layer, to be achieved by flying some form of fast-packet switch — this requires all signals received at the satellite to be demodulated and decoded, routed to the appropriate destination path and multiplexed with other traffic before re-coding and re-modulating for transmission,
- using the Ka-band (20—30 GHz) where up to 1 GHz of spectrum is available — the physics of the Ka-band means that smaller spotbeams can be generated than at Ku-band (given the same satellite antenna size) and this enables the spectrum resource to be reused several times over the same coverage area.

The first such systems are likely to be placed in GEO orbit with a basic architecture similar to that shown in Fig 3. Examples of such proposals include Spaceway from Hughes [6], Astrolink from Lockheed Martin [7] and Cyberstar from Loral [8]. Most systems propose providing global coverage using a constellation of 3—17 GEO satellites placed at strategic points around the world.

Ka-band VSAT antennas could be as small as 30 cm in diameter with a target price to the consumer of around \$1000. All systems aim to provide everything from basic voice service at 16—64 kbit/s to full motion interactive video at up to 8 Mbit/s, although support of broadcast or multicast video services is not yet well defined. User antenna

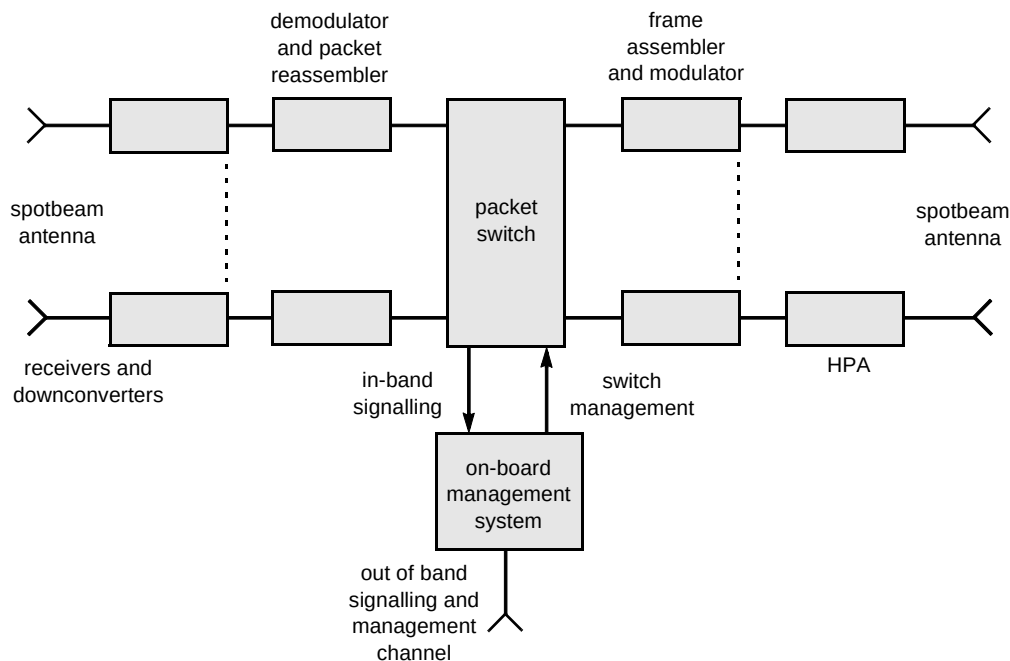


Fig 3 Basic architecture for a GEO fast-packet repeater.

size increases with required bit rate with the full 8 Mbit/s service requiring antennas of the order of 2–3 m. Anything bigger than 3 m would require a tracking mechanism which would have a great impact on the terminal price.

Space segment costs are predicted to be one to two orders of magnitude lower than the equivalent Ku-band capacity today because of the level of frequency reuse and the fact that the systems will be open access, rather than lease-based. However, the precise costs will be dependent on the fill-factor of the system which in turn is dependent on how the satellite capacity is managed. User tariffs are expected to be well under \$1 per minute for a duplex 64 kbit/s link. Licence costs will be key to the success of these systems and will need to be set at relatively low levels per terminal if licence charges are not to restrict the growth of Ka-band VSATs.

The system architecture and the service and network management philosophy are key issues. The system proposers use a variety of descriptions for their switching architecture — ‘ATM-like’ and ‘fast-packet’ being the most common. The choice of link-layer protocol will have a big impact on the ability of the operator or service provider to monitor quality of service (QoS) and control the grade of service (GoS) available to users. Support of network monitoring and management information, such as ATM OAM flows, will enable these systems to fully integrate with terrestrial systems using the same technology. There are two key questions for a service provider.

- Who will manage the switch capacity — the system operator or the service provider?
- Will a user be able to request a change in GoS on a per-call basis?

The answers to these questions will be closely linked with the commercial structure and architecture of these systems and this is still the subject of much public and private debate.

One characteristic of GEO-based satellite services that has a major impact on their use as an access technology is propagation delay due to satellite altitude. Interactive multimedia applications are expected to use the TCP/IP stack which has a maximum transmission window size of 64 kbit which can be sent before an acknowledgement is expected from the receiver. A GEO satellite at an altitude of approximately 36 000 km introduces a round-trip delay of about half a second and this limits the maximum throughput of a standard TCP session to 128 kbytes/s or 1 Mbit/s. This throughput is adequate for the majority of today’s applications but it will not support the interactive full-motion video applications of the future. Throughput can be increased through techniques such as window extension and selective acknowledgements but these options will not be available in the standard offering for some time.

System costs and performance can theoretically be improved further by the use of low-Earth orbit (LEO) systems operating at 700–1500 km altitude. Bringing the satellite closer to the Earth means that it is no longer stationary with respect to a fixed point on the Earth. Therefore a constellation of satellites is needed to provide 24-hour coverage of any location in the coverage area and each satellite ‘sees’ a smaller area. A LEO architecture can achieve several things:

- reduced power requirements for the radio link — this can be used to reduce antenna size or transmitted power from both the satellite and the user terminal,
- increased system capacity through each satellite reusing the available spectrum — this can bring down the cost per bit if the required fill is achieved,
- transmission delays similar to those of terrestrial networks — this means that standard protocols such as TCP/IP can be used to support applications.

These benefits are achieved at the cost of additional complexity in terms of traffic routing, call handover and network management. The network can be implemented in one of two ways — using transparent satellites with all network functionality located on the ground, or by flying switches on-board the satellites. Alcatel have proposed the SkyBridge [9] system of 64 satellites using the transparent architecture and Teledesic [10] (whose consortium includes Motorola, Boeing and Matra Marconi Space) has proposed a 288 satellite system which will fly some kind of switch. SkyBridge also differs from the other broadband systems in that it will operate in the Ku-band and will need to avoid interfering with other systems operating in the same band. User costs for both systems are expected to be of the same order as those for a GEO solution.

Placing a switch on-board any satellite raises questions over reliability due to the susceptibility of semiconductor elements, such as memories, to faults caused by cosmic radiation. The degree to which this can be mitigated will drive the final system architecture, functionality and lifetime.

4. Mobile communications services

4.1 Current mobile systems

The evolution of mobile satellite technology has so far been dominated by the International Maritime Satellite Organisation (Inmarsat). The Inmarsat system was originally designed to provide service to the maritime community and to niche markets that required transportable communications in remote parts of the world, e.g. mineral exploration companies. Both of these markets could accept the relatively large user terminals (of the order of 1 m diameter antennas) associated with early systems in order to

achieve basic connectivity. Since Inmarsat was created the cost of satellite technology has come down and spacecraft and launchers have become more reliable. Hence, several other systems have been (or are about to be) launched around the world to serve regional markets or specific applications. Table 1 provides a summary of the services supported by the Inmarsat system on a global basis. All except the Aero service can and have been used to provide communications to semi-fixed terminals in remote areas or for emergency services where terrestrial systems have been disrupted.

Inmarsat is in the process of procuring their next generation of satellites called the Horizons system. Horizons will use spectrum in the 2 GHz band to support bearer services up to 144 kbit/s. The aim is to provide the satellite component of IMT2000/UMTS. System capacity will be limited by the available spectrum and service costs are likely to reflect this and therefore will not represent a threat to terrestrial ISDN services.

Inmarsat satellites use the GEO orbit with the associated limitations on the radio power budget and transmission delay. New systems which plan to start operation over the next two years use LEO or medium Earth orbit (MEO). MEO orbits range roughly between 10 000 and 15 000 km. The spacing of these nominal orbital altitudes is due to the Van Allen radiation belts which would damage the satellites and their payloads. These systems are called global mobile personal communications systems (GMPCSs).

4.2 Global mobile personal communications systems

Three major players have emerged in the GMPCS handheld market — ICO Global Communications, Iridium and Globalstar [11]. Each system has its own characteristic system architecture, orbits, technologies and approach to local service provision. The capacity of these satellite systems is very small compared with terrestrial cellular and poses more opportunity than threat to national cellular network operators and service providers.

Coverage of satellite systems is limited in urban and indoor environments because of tight power budgets and terrestrial clutter shadowing the radio path — this narrows target applications to certain niche markets such as international business travellers, cellular extension into remote locations and areas where communications infrastructure has yet to be developed. In each case, global operation is the prime requirement.

All of these systems are designed to be used with mobile terminals of approximately the same size and weight as current terrestrial cellular telephones supporting 4.8 kbit/s voice and 2.4 kbit/s data. Most of the consortia proposing these services intend to lease capacity on their systems to non-common carriers for resale to customers rather than acting as sole vendors of the services themselves. Cellular service providers in the USA and elsewhere are looking to these systems as a way to extending their coverage for roaming customers. A simple summary of the systems is shown in Table 2. Note that the system capacity may not be

Table 1 Summary of Inmarsat mobile satellite services.

System	Inmarsat-A	Inmarsat-B	Inmarsat-M/mini-M	Inmarsat-C	Inmarsat-Aero
Type	Analogue voice digital data	Digital	Digital	Digital	Digital
Services and channel rates	Circuit switched Voice: 15 kHz FM Data: up to 9.6 kbit/s and 64 kbit/s Fax: up to 9.6 kbit/s Standard telex	Circuit switched Voice: 9.6—16 kbit/s Data: 9.6 kbit/s and 64 kbit/s Fax: 9.6 kbit/s Standard telex	Circuit switched Voice: 4.8 kbit/s Data: 2.4 kbit/s Fax: 2.4 kbit/s	Store and forward Data: 600 bit/s	Circuit switched Voice: 4.8—9.6 kbit/s Data: 2.4 kbit/s Packet data Fax: 2.4 kbit/s
User terminal	0.3—1.5 m steerable and/or gyro-stabilised parabolic antenna	20—150 kg steerable parabolic antenna and flat-plane arrays	Steered or gyro-stabi- lised reflectors to flat- plane arrays down to laptop PC size (2—3 kg)	Omni' antenna, laptop size DTE	Gyro-stabilised air- craft mounted reflec- tor, handset/jack/ keypad interface

Table 2 Simple summary of the GMPCS proposals .

System name	Orbit	Operational satellites	System call capacity	Access technology	Complexity	Full service
ICO	MEO	10	45 000	TDMA	Medium	2000
Iridium	LEO	66	140 000	TDMA	Very high	1998
Globalstar	LEO	48	92 500	CDMA	High	1999

fully utilised because not all of it can be focused over land all of the time.

All these systems have required substantial investment (in the region of \$2.5—3.5bn) and are targeting essentially the same markets. Call costs will range from about \$4 per minute end-to-end for Iridium to around \$0.6 per minute for air time only from Globalstar. It is worth noting that while these systems were initially targeted at mobile communications, about two-thirds of their market is expected to be for fixed terminals in developing countries. The ability of these systems to focus capacity is limited primarily by the available spectrum and the number of satellites visible from the ground and so these systems are not suitable for high-density mass-market solutions. However, GMPCS systems can provide expedient access in areas of poor infrastructure such as Africa or China until terrestrial alternatives are available. This fixed market is likely to come under significant threat from the broadband satellite systems which will be able to provide similar (and more advanced) services to fixed terminals at a much lower cost. The 20/30 GHz band is considered unsuitable for true mobile applications at this time because the Ka-band mobile channel is not well understood.

Despite the global reach of mobile satellite systems they are unlikely to ever pose a threat to terrestrial cellular due to their limited capacity and high infrastructure costs. Available spectrum is limited — about 60 MHz at L-band and a further 60 MHz at 2 GHz in the UMTS bands. Although technology will enable the cost-per-bit to fall over time, it is unlikely to ever catch up with terrestrial or cellular system costs.

5. Conclusions

To date, satellite services have not been widely deployed in the access domain for interactive services. They have been limited to broadcast TV for residential customers and niche applications within the business markets. More emphasis has been placed on their ubiquitous coverage capability and their ability to provide service where all other technologies fail.

Satellite delivery is likely to remain popular for pay TV even in the face of competition from the digital terrestrial TV services which are planned to start before Christmas this year because it offers much more capacity for the introduction of near video-on-demand services and HDTV in the future. It provides a cost-effective means of delivering business TV and data services both in the UK and overseas. It is also likely to become used for push-Internet services initially to PCs and LANs but in time also to TVs via set-top boxes. The use of satellites for interactive TV return channels and for fast pull-Internet delivery is less

easy to predict. There will for instance be little benefit in introducing a satellite return channel in the UK if the PSTN-based solution offered by BIB (which should provide a much higher availability and data rate) is accepted by the public. The attraction of this service and of satellite-delivered fast pull-Internet will also be further reduced when BT is able to introduce broadband network-based services via xDSL at an acceptable price.

Current VSAT systems are too expensive to consider for basic access services due to the high cost of both the user equipment and air time. VSAT is a good single-platform solution for international private networks for MNCs and can be used to fill gaps in a company network where terrestrial technologies are not viable. New broadband satellite systems will initially target the same sectors as existing VSAT systems from about 2002. A high system-fill factor could bring usage costs down to the point where satellite becomes a viable option for SME and high-end residential access services. These platforms could also 'trail blaze' in developing regions where a critical mass of customers needs to be established before it is economic to roll out terrestrial infrastructure for basic services. There are significant technology challenges to overcome in the provision of switches on-board satellites if these systems are to succeed, particularly with respect to capacity management. Mobile satellite systems support a variety of niche and specialised markets. Current cost and capability limitations mean that they are not viable as an access solution for densely populated areas. The introduction of GMPCS systems in late 1998 will herald the start of a new era in satellite communications as satellite system user equipment size and functionality will move closer to their terrestrial equivalent.

The goal of all the new satellite systems being deployed over the next 5 years is to drive down the cost of satellite access so that more users can take advantage of their global connectivity. This should ensure that satellite solutions become an integral part of the global service providers' portfolio in the next millennium.

References

- 1 Dobbie W H: 'Services, technology and standards for broadcast-related multimedia', *BT Technol J*, 13, No 4, pp 44—54 (October 1995).
- 2 Mills G et al: 'DVB specifications for broadcast-related interactive TV services', *Electronics & Communications Engineering J*, IEE, pp 38—42 (February 1997).
- 3 Dobbie W H and Bissell R A: 'Interactive TV services, standards and set-top box evolution', *International Symposium on Services, and Local Access*, Venice (March 1998).

- 4 Fitch M: 'Network management and services using ATM over satellite', Presented at the IEE Colloquium on Current Developments in INTELSTAT, IEE publication reference 1997/367 (October 1997).
- 5 UK DTI Radiocommunications Agency World Wide Web pages at <http://www.open.gov.uk/radiocom/rahome.htm>
- 6 Hughes Communications Galaxy, Inc, Application before the Federal Communications Commission for Spaceway: A Global Network of Ka-Band Fixed Communications Satellites (July 1994).
- 7 Lockheed Martin Corporation, Application before the Federal Communications Commission for the Astrolink Communications Satellite System (September 1995).
- 8 Loral Aerospace Holdings, Inc, Application before the Federal Communications Commission to Construct, Launch and Operate a Communications Satellite in the Domestic Fixed-Satellite Service (May 1995).
- 9 SkyBridge LLC World Wide Web pages at <http://www.skybridgesatellite.com/>
- 10 Teledesic Corp World Wide Web pages at <http://www.teledesic.com/index.html>
- 11 Finean R J: 'Mobile satellite communications proposals', BT Technol J, 14, No 3, pp 74—80 (July 1996).



a Member of the IEE.

Jon Wakeling has worked in the communications industry for twenty years. He joined BT Laboratories in 1989 after graduating from Plymouth Polytechnic with a degree in communications engineering. He has specialised in satellite communications systems with a particular focus on mobile satellite system performance enhancement and Ka-band system technology. He is currently working on the assessment of future broadband satellite systems and related network management requirements. He manages the Satellite Systems team at BT Laboratories. He is a Chartered Engineer and



Broadcasting Project Technical Module.

Will Dobbie received a PhD from Essex University in 1990. He worked in the Royal Air Force and Marconi Communications Systems Ltd before joining BT in 1982. His background is in bandwidth efficient satellite TV and data transmission systems. More recently he has become involved with the specification of end-user equipment for new interactive multimedia services, particularly those linked to broadcast services. He is currently a technical advisor on interactive TV within the products and services engineering centre at BT Laboratories. He represents BT in the European Digital Video